COMPUTING INFRASTRUCTURES

Week 1: Intro to Infrastructure

Exploring the historical evolution from Mainframes to Cloud Ecosystems and establishing practical local discovery audits.

PHASE 01 — THEORETICAL FOUNDATIONS

The History & Evolution

From Mainframes to Minicomputers

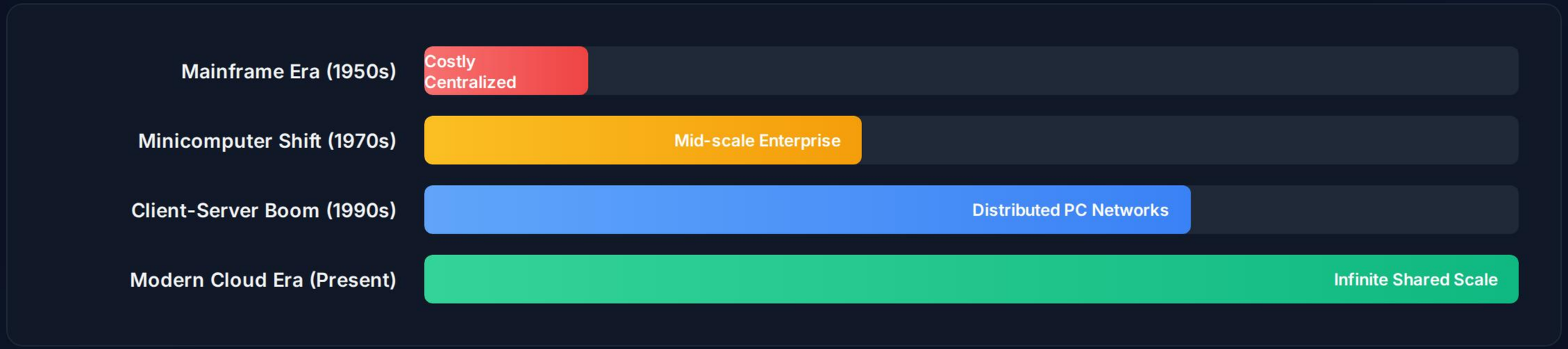
The Centralized Mainframe Era

Starting in the 1950s, computing power resided on massive, physical host machines (e.g., IBM System/360). Access was completely centralized, and users interacted with high-performance mainframes using dummy terminals via batch-processing techniques.

The Decoupled Minicomputer Shift

Introduced in the late 1960s and 1970s, minicomputers (e.g., DEC PDP-11) decentralized computing for mid-sized business operations. This era pioneered multi-tasking Unix Operating Systems and democratized accessibility at significantly reduced costs.

Exponential Scale & Cost Shifts



Visual comparison illustrating the evolution from isolated highly expensive physical compute nodes to modern on-demand utility-based resource distribution.

Decentralization & Cloud Ecosystem



Client-Server Model

The 1980s and 90s saw computing disperse to individual PCs. Dedicated servers provided database, storage, and print resources to connected client workstations across localized networks.



Grid & Clusters

The early 2000s grouped multiple lower-cost systems into tight software networks. These coordinated setups achieved high fault tolerance, redundant processing power, and massive parallel throughput.



The Cloud Utility

Modern paradigms virtualize underlying hardware arrays. Infinite processing capacity, storage blocks, and logical layers are provisioned instantaneously via utility payment schemes (AWS, Azure).

Defining the Modern Stack

The structural system layers underpinning modern compute operations:

- **Physical Hardware:** High-density Rack servers, scalable dynamic RAM, NVMe arrays, and specialized processors (CPUs/GPUs).
- **Network Fabrics:** Top-of-rack leaf/spine switches, physical routing paths, DNS backbones, and optical links.
- **Storage Systems:** Consolidated Storage Area Networks (SANs), Network Attached Storage, and structured databases.
- **Virtual Orchestration:** Hypervisors dividing bare-metal capabilities into modular guest virtual operating engines.



PHASE 02 — PRACTICAL APPLICATION

Laboratory: Local Machine Audits

System Discovery on Windows OS

Windows native Command Prompt (cmd) / PowerShell utilities enable quick hardware auditing:

- **systeminfo:** Returns full OS details, CPU model, BIOS versions, hotfixes, and physical RAM capacities.
- **wmic cpu get name:** Query direct WMI interfaces for raw hardware, logic cores, and manufacturer tags.
- **dxdiag /t:** Direct diagnostic exporter listing graphic capabilities and display architectures to text files.

Sample Windows Powershell Audit:

```
# Get total physical memory, CPU status
Get-CimInstance Win32_ComputerSystem |
Select-Object Name, TotalPhysicalMemory
# Retrieve operational disk partitions
Get-PhysicalDisk | Select-Object FriendlyName, Size
```

System Discovery on Unix / Linux

Linux CLI Hardware Commands:

```
# Check complete kernel and hardware spec
$ uname -a

# Gather total physical hardware layout
$ sudo lshw -short

# Display CPU architecture capabilities
$ lscpu

# Read live dynamic RAM allocation
$ free -h
```

Core Unix-based hardware reporting concepts:

- **uname:** Instantly displays OS Kernel releases, release names, and architecture variants (e.g., x86_64, arm64).
- **lshw / lscpu:** Interrogates low-level hardware structures inside Unix systems to map logic configurations.
- **The /proc directory:** A virtual file system mapping operational processes to live virtual memory spaces.

Lab Protocol & Audit Objectives

Laboratory 01 Step-by-Step Deliverables:

- **Execution:** Access the local CLI tool according to your host operating system (CMD, PowerShell, or Bash).
- **System Profiling:** Execute terminal queries to record the CPU Model, active cores, and total RAM capacity.
- **Network Profiling:** Identify target IPv4, loopback interface configurations, and hardware MAC endpoints.
- **Data Export:** Redirect dynamic screen output into organized local technical summary text files.



OS Command-Line Comparison Matrix

Target System Metric	Windows Command (CMD/PS)	Linux Native Utility	macOS Utility (Darwin)
OS Version / Kernel	`systeminfo | findstr OS`	`uname -a`	`sw_vers` or `uname -r`
CPU Information	`wmic cpu get name`	`lscpu`	`sysctl -n machdep.cpu.brand_string`
Active Memory Info	`systeminfo | findstr RAM`	`free -h` or `vmstat`	`vm_stat`
Network Interfaces	`ipconfig /all`	`ip link show` or `ifconfig`	`ifconfig -a`

WEEK 1 INTERACTIVE WRAP-UP

Discussion & Lab Delivery

Complete the Lab 01 report template using raw output captured directly from your local terminal session. Submit standard Markdown or PDF files by the designated lecture deadline.

→ NEXT WEEK: ENTERPRISE SERVER HARDWARE CONFIGURATIONS

Image Sources



https://static.vecteezy.com/system/resources/previews/069/780/218/large_2x/close-up-of-modern-server-rack-with-glowing-blue-and-green-lights-in-data-center-photo.jpeg

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